Proc. of the 9th International Symposium on Geotechnical Safety and Risk (ISGSR)
Edited by Zhongqiang Liu, Jian Dai and Kate Robinson
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doi: 10.3850/GRF-25280825_isgsr-207-P254-cd

CALIBRATION OFPARTIAL SAFETY FACTOR FOR CPT-BASED AXIAL PILE CAPACITY DESIGN METHODS

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In addition to the API-method, the API RP 2GEO (2011) and ISO 19902 (2007) guidelines include four CPT-based methods for calculating the axial capacity of piles, namely the Fugro, ICP, NGI and UWA methods. However, the values of partial safety or resistance factors to be used with the CPT-based methods are not given. The designer is required to select an appropriate safety factor when using the CPT-based methods. The difficulty lies in deciding which safety factor or combination of partial safety factors will ensure a consistent safety level for different soil conditions and pile dimensions. In a recent Joint-Industry Project, a reliability-based approach was adopted to calibrate the resistance factor of the four CPT-based methods with 13 case studies. The goal of this case-by-case calibration was to ensure that the annual failure probability of the designed pile foundation was less than target probabilities of 10^{-3}, 10^{-4} or 10^{-5}, and to compare the results with the safety level implied by the current API-pile design method. The new code calibration method sets two working hypotheses. The idea behind these hypotheses is to base the calibration not only on the computed annual failure probabilities for each pile design method but also on the comparison of the predicted axial capacities with each method for the individual case studies relative to each other. The paper explains the two working hypotheses and describes the approach adopted for the calibration and optimisation of the partial safety factor. The new code calibration approach gave consistent results for the partial safety factor for pile design with the CPT-based methods. The paper makes recommendations for the resistance factors to be used with the four CPT-based axial pile capacity design methods in the API and ISO guidelines.

Keywords: pile foundations; reliability; calibration; resistance factor; failure probability.

1. Introduction

The design of tubular steel pile foundations for offshore installations has traditionally been based on the estimation of axial pile capacity using the so-called API-method. There is general consensus that the API-method underpredicts the pile capacity in sand profiles, especially in dense sand. The API method is believed to give reliable results for piles in clays that are similar to the Gulf of Mexico clays.

Inthe past two decades, new methods for predicting the axial pile capacity from the cone penetration tests (CPT) have been proposed. The API RP2A (RP2GEO) (2011) and ISO 19902 (2007) guidelines include four so-called CPT-methods for calculating the axial capacity of piles in sands, namely the Fugro, ICP, NGI and UWA methods. The CPT-based methods are believed to be better suited than the API-method to predict the pile capacity for piles in sand. The guidelines require that the designer selects an appropriate safety factor when using the CPT-based methods. The difficulty lies in deciding which safety factor will ensure a consistent safety level and deciding of a safety factor applicable for different soil conditions and pile dimensions. With newer design methods, it is not uncommon to apply a higher resistance factor on the ultimate axial pile capacity than the resistance factor used with the API method. This can have dramatic consequences on the required pile penetration depth.

The goal of the study was to recommend the 'appropriate' resistance factor forthe design of the piles. This goal was achieved through a reliability-based approach. A reliability approach enables one to achieve systematic treatment of the uncertainties associated with environmental loadings and geotechnical conditions and models and to circumvent the use of safety factors.In recent years, reliability-based approaches have gained momentum for the management of the safety of offshore structures and other infrastructure (Lacasse et al., 2019; Nadim et al., 2022; Liu et al., 2022: Lacasse and Höeg, 2019).

2. Pile capacity design methods

Five methods for the calculation of the axial pile capacity were included in the reliability study: the current API method, the simplified ICP-05 method, the NGI-05 method, the Fugro 95-10 method (in the initial calculations, the Fugro-05 method for clay was considered. But this method was replaced by the newer Fugro-10 method in the final analysis), and the UWA-05 method. Each of the methods are described in the API RP2GEO (API, 2011) and ISO 19902 (ISO, 2007) guidelines. In addition, the UWA-13 method for clays is included.

3.Calibration of resistance factors

The goal of the calibration analysis is to assess the combinations of partial safety factors that would ensure that

the annual probability of foundation failure is less than a target value in a load and resistance factor design(LRFD) format. The design inequality, used in the case studies for calibration of the partial resistance factors, has the following format:

$$[\gamma_{l\,stat} \bullet P_{sta} + \gamma_{l\,env} \bullet P_{env}] \;\; d \;\; Q_{ult}/\gamma_m \qquad (1)$$

where $\gamma_{l\,stat}$ = load factor on the representative value of the static load
$\gamma_{l\,env}$ = load factor on the representative value of the environmental load
γ_m = resistance factor on the ultimate pile capacity
P_{sta} = representative value of the static load
P_{env} =representative value of the environmental load (depends on limit state analysed)
Q_{ult} = ultimate pile capacity under the applied loads

Figure 1 illustrates the procedure with a two-dimensional simplification of the overlap of the probabilistic ultimate pile capacity (Q_{ult}) and probabilistic peak annual axial load (P_{stat} +P_{env}). The shape of the PDF for (P_{stat} +P_{env})is taken to be the same as it is shifted to the right. P_{f1} represents the annual failure probability obtained with the mean and standard deviation of all random parameters and P_{f2} represents the target annual probability of failure (i.e. P_{f1}= 10^{-4}/yr in Fig. 1). The calibration is done by applying a scaling factor on the loads from the calculated P_f to the target annual probability of failure. The design point is the vector of load and capacity combinations leading to failure and is denoted with a star (*) in the procedure. To draw general conclusions on the partial safety factors to be used with each of the CPT-based methods, the study included 13 case studiesfrom six different countries from three continents.The13 case studies belonged to two groups: (I) Piles with tip in SAND, with no or thin layers of highly overconsolidated clay. with low-medium plasticity (7 case studies, whereof 3 cases could be said alternating layers of clay and dense sand):(II) Piles with tip in CLAY, mainly overconsolidated clay of low to medium plasticity, with some medium dense to dense sands(6 case studies).

The resistance factors were calibrated for three target annual probabilities of failure, 10^{-3}/year, 10^{-4}/year and 10^{-5}/year.

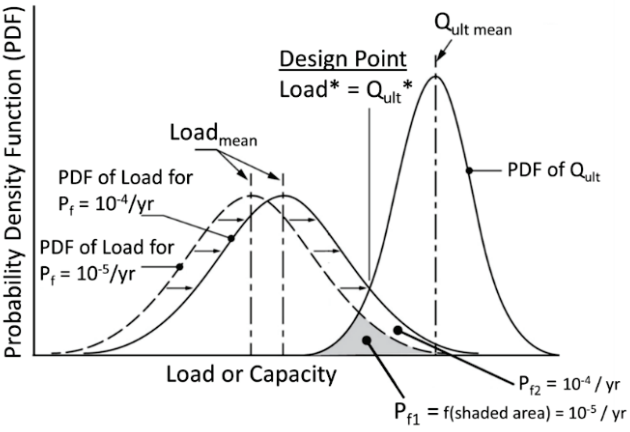


Fig. 1.2D simplification of the PDFs of the load (P_{env} + P_{stat}) and ultimate pile capacity Q_{ult} (P_{f1} and P_{f2} volumes are not exact, they are for illustration purpose only)

4.Working hypotheses for code calibration

Depending on the pile design method used, and the approach adapted for modelling uncertainty, the calculated annual foundation failure probability for each case study can vary over a very large range for a given pile penetration. In the calibration and optimisation exercise, two working hypotheses were made to help reduce the variability of the results. The basic idea behind these two working hypotheses was to base the calibration not only on the computed annual failure probabilities for individual methods, but also on the comparison of the predicted axial capacities with different methods for individual case studies relative to each other.

The two working hypotheses that were used in the calibration and optimisation exercise were:

- **Working Hypothesis 1:** With a given pile geometry in one location, and for given static and environmental axial loads, the methodthat predicts the lowest failure probability is the most robust method for estimating the axial pile capacity for that pile geometry at that location and the calculated failure probability with that method is the most realistic estimate of the actual failure probability.
- **Working Hypothesis 2:** The method identified by Working Hypothesis 1 for a specific case study is denoted Method A. Method A requires a resistance factor equal to γ_m^A in a deterministic design to achieve

the target annual failure probability at that site (Working Hypothesis 1). If another method, denoted Method B, predicts a lower deterministic axial capacity than Method A at the same design pile penetration depth, then Method B combined with resistance factor γ_m^A will achieve a design that has a lower annual failure probability than the target value.

Based on the two conjectures above, the calculated failure probabilities and calibrated resistance factors from the case studies were adjusted with the following procedure:

- The smallest annual failure probability (or highest annual reliability index β) evaluated with any of the methods is considered to be the most realistic annual failure probability (or annual reliability index) of that case. The method that gives the smallest annual failure probability (or highest annual β) is denoted Method A. This is regarded as the "actual value" for each of the case studies.
- The smallest value of the calibrated γ_m for one given design method is assumed to be the realistic value for each case study.
- If another method (denoted Method B) predicts a lower deterministic axial bearing capacity than Method A at a site, the resistance factor for Method B is set equal to the resistance factor calibrated for Method A.
- If Method B predicts a higher deterministic mean capacity than Method A for a given pile penetration depth, the resistance factor for Method B is set to the smallest value of γ_m^B (calculated value of resistance factor for Method B and $\gamma_m^A \times$ (Axial capacity of Method B) / (Axial capacity of Method A).

5.Optimization procedure and calibration of resistance factor

The calibration analysis was done by using the Lind (1977) penalty function(Fig.2). This approach penalises under-design more than overdesign. The Lind penalty function is defined as:

$$p = k\,(\beta - \beta_t) + exp[-k(\beta - \beta_t)] - 1 \quad (2)$$

where k is a constant, β_t is the target reliability index and β is the reliability index obtained with a given value of resistance factor γ_m for the case study in question. The value of γ_m is varied over the range of the obtained results for all cases and for each γ_m, the sum of the penalties for all cases is calculated. The optimum value of γ_m is the value that gives the minimum total penalty. Ronold (1999) used k = 1.5 for optimization of a design code for tension piles. He alsodid a study of the sensitivity of the results to coefficient k in Eq. 2, and suggested values between 0.8 and 5 did not have a significant effect on the results.

In this study, a value of k = 3.0 was assumed and the optimum resistance factors for target reliability index β corresponding to annual *probabilities* (P_f) of 10^{-5}, 10^{-4}, 10^{-3} were calculated. A sensitivity study was made with k between 1,0 and 4.0, and no significant changes in the results were noted.

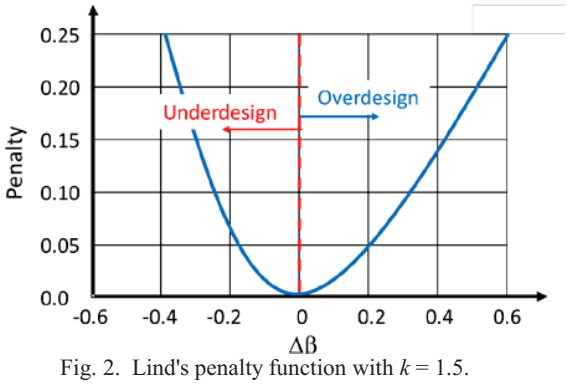


Fig. 2. Lind's penalty function with k = 1.5.

6. Optimisation of resistance factors for CPT-based methods

6.1. *Optimised resistance factors*

The results obtained with the optimisation exercise are quite consistent, and can be summarised as follows:

Target annual $P_f = 10^{-3}$: γ_m = 1.30 for NGI, ICP and UWA methods, = 1.40 for Fugro method
Target annual $P_f = 10^{-4}$: γ_m = 1.50 for NGI, ICP and UWA methods, = 1.60 for Fugro method
Target annual $P_f = 10^{-5}$: γ_m = 1.70 for NGI, UWA methods, = 1.80 for ICP and Fugro method

These resistance factors are with reference to the axial capacity evaluated with the mean estimate of the soil properties and cone resistance used in the calculations of axial pile capacity.

6.2.*The implied target reliability level for the API-method*

In order to assess the implicit target reliability level in the current API-design, the penalty function approach of Lind (Eq. 2) was applied to the results of case studies to answer the following question:

> "Given that pile design according to the API guidelines results in the annual failure probabilities and reliability indices for the cases considered, what value of target annual reliability index $\beta_t$did the guideline implicitly aim to achieve?"

When the Working Hypothesis 1 was used, a **target annual failure probability between $5{\cdot}10^{-5}$ and 10^{-4}** is achieved (annual β_t between 3.7 and 3.9). This result is consistent with acceptable safety levels in the industry and experience with piles designed with the API-method.

6.3.*Recommended resistance factors for axial pile capacity design*

These recommended resistance factors are the optimised values listed in Section 6.1 increased by about 5%, to account for the increase in capacity with pile diameter due to the analytical formulation for the ICP, UWA and Fugro CPT-based methods.The recommended resistance factors in Table 1 are with reference to the mean axial capacity obtained with the mean value of soil parameters and cone resistance.The other parameters in the design inequality have the following values: $\gamma_{l\ stat}$= 1.0, $\gamma_{l\ env}$= 1.3, $P_{env}^{\ char}$=Characteristic environmental load with return period of 100 years for oil and gas platforms.

Table 1. Recommended resistance factors for three target failure probabilities and resistance factors for achieving the safety level implied by the API-method for the four CPT-based methods.

Method	γ_m for target $P_{f,\ annual}$= 10^{-3}	γ_m for target $P_{f,\ annual}$= 10^{-4}	γ_m for target $P_{f,\ annual}$ = 10^{-5}	γ_m for achieving implied$P_{f,\ annual}$ by API
NGI-05	1.35	1.6	1.8	1.65
ICP-05	1.35	1.6	1.9	1.70
Fugro-05/10	1.45	1.7	1.9	1.75
UWA-OS/13	1.35	1.6	1.8	1.65

Acknowledgement

The work was part of the JIP "Reliability of axial pile capacity calculation methods" carried out by NGI and UWA. The JIP was sponsored by Equinor AS, Lundin Norway AS, Ørsted, DNV·GL AS, ONGC and Petrobras. The technical and financial support from the sponsors is gratefully acknowledged. Preparation of the manuscript was also supported by the basic funding to NGI from The Research Council of Norway.

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